MACHINE LEARNING CANVAS

Designed for: | Designed by: | Date: | Iteration:

PREDICTION TASK

What is the type of task?
Which entity are predictions made on?
What are the possible outcomes to predict?
When are outcomes observed?

DECISIONS

How are predictions turned into actionable recommendations or decisions for the end-user? (Mention parameters of the process / application for this.)

VALUE PROPOSITION

Who is the end beneficiary, and what specific pain points are addressed?
How will the ML solution integrate with their workflow, and through which user interfaces?

DATA COLLECTION

How is the initial set of entities and outcomes sourced (e.g., database extracts, API pulls, manual labeling)?
What strategies are in place to update data continuously while controlling cost and maintaining freshness?

DATA SOURCES

Where can we get data on entities and observed outcomes? (Mention internal and external database tables or API methods.)

IMPACT SIMULATION

What are the cost/gain values for (in)correct decisions?
Which data is used to simulate pre-deployment impact?
What are the criteria for deployment?
Are there fairness constraints?

MAKING PREDICTIONS

Are predictions made in batch or in real time?
How frequently?
How much time is available for this (including featurization and decisions)?
Which computational resources are used?

BUILDING MODELS

How many models are needed in production?
When should they be updated?
How much time is available for this (including featurization and analysis)?
Which computation resources are used?

FEATURES

What representations are used for entities at prediction time?
What aggregations or transformations are applied to raw data sources?

MONITORING

Which metrics and KPIs are used to track the ML solution's impact once deployed, both for end-users and for the business?
How often should they be reviewed?

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What your canvas is for

A filled canvas does 4 jobs, and each one needs certain boxes filled in with care.

- **Deciding what's worth building.** Put each idea on its own canvas and you can compare the ideas on the same terms: what each could gain in a year, and whether the outcomes a model would learn from were ever recorded. That takes dollar amounts in the Impact Simulation box, and a named person in the Value Proposition box who acts on each prediction.
- **Agreeing on what gets built.** Some of the choices that set a system's value look technical, but they're business decisions. What counts as a lost customer: 60 days without buying, or 90? How far ahead do you predict? How many flagged cases can your team act on? On the canvas, business, data and engineering make these choices together, instead of leaving them to whoever writes the code.
- **Knowing when the model is good enough.** The Impact Simulation box sets the performance the model has to reach to pay off. Test the model on enough of the cases you're predicting: 35,000 machine-hours of data can hold just 3 failures, too few to measure anything.
- **Learning after launch.** If acting on a prediction changes the outcome, keep a small share of flagged cases untouched and record that share in the Decisions box, so you can see what your action changed.

Whatever you can't answer is the work left before anyone writes code.

"The Canvas allowed us to structure and focus our delivery sprints. The ML system went live in June 2020 and still is."
— Christian Schiller, Senior IT & Enterprise Architect, Deutsche Telekom

"MLC played an essential role in a positive project outcome. It is now being used throughout the organization."
— Daitan, now part of Coforge, a global IT services firm

Book a project review

Louis Dorard created the canvas in 2015. Book a free 30-minute call, then send him your canvas draft, or a few lines about the project if you haven't started one. He prepares the call from what you send, and you'll leave knowing where the project is likely to break, and what to check before anyone builds it.

Book a 30-minute call →

Reference the Machine Learning Canvas

If you need to reference the Canvas formally (e.g. in documentation or publications), you may use:

```
@misc{mlc,
  title  = "Machine Learning Canvas",
  author = "Louis Dorard (OWNML)",
  url    = "https://www.ownml.co/",
  year   = "2015"
}
```